Monday Warm Up: Unit 7: Forces V

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1 Memory Bank

- $\hat{i}\cdot\hat{i} = 1$, $\hat{i}\cdot\hat{j} = 0$ ... The dot product, unit vectors
- $\vec{u}\cdot\vec{v} = uv\cos\theta$ ... The dot product of two vectors is the product of the magnitudes times the cosine of the angle between them.
- $\hat{i}\times\hat{j} = \hat{k}$, $\hat{k}\times\hat{i} = \hat{j}$, $\hat{j}\times\hat{k} = \hat{i}$ ... The *cross-product*, unit vectors
- $\hat{j}\times\hat{i} = -\hat{k}$, $\hat{i}\times\hat{k} = -\hat{j}$, $\hat{k}\times\hat{j} = -\hat{i}$ ... When the unit vectors are not in alphabetical order, there is a minus sign.
- $\tau = \vec{r}\times\vec{F}$ ... Definition of torque.
- $|\tau| = rF\sin\theta$ ... The magnitude of the torque is the product of r and F, times the sine of the angle between them. The distance r is between the force and the pivot point.
- Torque is the angular version of force. It is the result of a force F, separated from a pivot by a distance r, that rotates the system about the pivot. The angle between the force F and the distance r is θ.
- **For sytems in static equilibrium, (1) the net force is zero, and (2) the net torque is zero.**

2 Dot Product, and Cross Product

1. Calculate the dot product:
   - $\vec{a} = 2\hat{i} - 2\hat{j}$, $\vec{b} = -4\hat{i} + 4\hat{j}$.
   - $\vec{a} = 2\hat{i}$, $\vec{b} = -4\hat{k}$.
2. Calculate the cross product:
   - $\vec{a} = 2\hat{i}$, $\vec{b} = -4\hat{k}$.
   - $\vec{a} = 2\hat{i} + 2\hat{j}$, $\vec{b} = -2\hat{i} - 2\hat{j}$.

3 Statics

1. A device for exercising the upper leg muscle is shown in Fig. 1, together with a schematic representation of an equivalent lever system. Calculate the magnitude of the force exerted by the upper leg muscle to lift the mass at a constant speed.

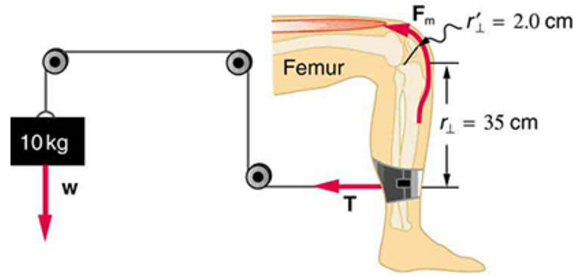

Figure 1: The quadriceps provide torque.

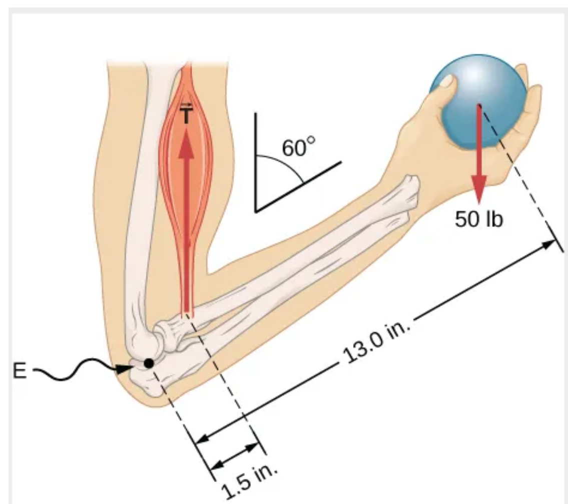

Figure 2: A balanced system.

2. Consider Fig. 2. The tension in the biceps acts along the vertical direction given by gravity. What tension must the muscle exert to hold the forearm at the position shown? Assume that the forearm's weight is negligible.

3. What is the force on the elbow joint?